

Empirical Research Methods

— Assignment 6 —

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Foreword In this exercise we evaluate the impact of a reform. The aim of the exercise is to understand difference-in-differences estimation. We will use the data set `careworkers_did.csv`. The data set contains information on 1997 individuals aged 19-40 from two countries, A and B, from 1991 to 1995. The president of Country A is concerned that, given the country's demographic trends, society may face a shortage of care services because the number of care workers is too low. In 1993, a reform was introduced in country A to increase the number of careworkers. The reform relieved all individuals who worked as careworkers from paying taxes entirely. We are interested in evaluating this reform.

Data Set: `careworkers_did.csv`.

Table 1: Description of Variables

Variable name	Description
<i>ID</i>	Town identifier
<i>COUNTRY</i>	Country of the town (A or B)
<i>YIAR</i>	Year of observations
<i>CAREWORKERS</i>	Number of careworkers per hundred individuals living in a town

Part 0 First, prepare the data for the analysis.

1. Import the dataset `careworkers_did.csv`, explore it, and clean it if necessary. Make sure it is plausible (no negative values, correct variable names). **[0.25 point]**

Part I In this part, we will check whether a difference-in-differences estimation is the appropriate strategy to estimate the causal impact of the reform on the number of careworkers.

1. Generate the following variables for use later in the analysis:
 - **POST**: a dummy variable that takes the value 1 if the year of observation is after 1992, and 0 otherwise **[0.25 point]**
 - **TREATED**: a dummy variable that takes the value 1 if the town is in country A, and 0 otherwise. **[0.25 point]**
2. What is the key assumption of the difference-in-differences estimation? (1 sentence) **[0.25 point]**
3. Provide visual evidence for the assumption. **[0.5 point]**
4. Provide formal evidence for the assumption explicitly testing it with a regression. **[0.5 point]**
5. In light of the two prior answers, is the assumption plausible in this context? (1 sentence) **[0.5 point]**
6. What is a potential concern in the structure of the data? (1 sentence) **[0.25 point]**

Part II In this part, we will estimate the impact of the reform on the number of careworkers using a difference-in-differences estimation.

1. Estimate the following difference-in-differences regression model:

$$CAREWORKERS_{it} = \beta_0 + \beta_1 POST_{it} + \beta_2 TREATED_i + \beta_3 POST_{it} \times TREATED_i + \epsilon_{it}$$

where ϵ_{it} is the error term.

2. Which one is your main coefficient of interest and why? (1 sentence). **[0.5 point]**

3. Interpret the main coefficient of interest. (1 sentence). [**0.25 point**]
4. What is the interpretation of the other coefficients? Form one sentence per coefficient (you should have 3 sentences). [**1 point**]
5. Was the reform successful in increasing the number of careworkers? (1 sentence) [**0.5 point**]